CAM OVERFLOW ATTACK

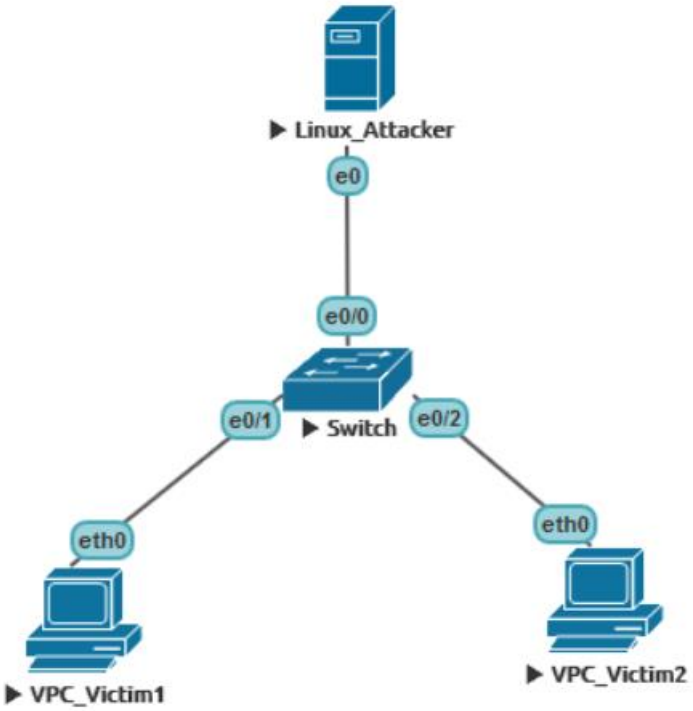
THE GOAL

A forwarding information base (FIB), also known as a forwarding table or CAM table, is most commonly used in network bridging, routing, and similar functions to find the proper interface to which the input interface should forward a packet. It is a dynamic table that maps MAC addresses to ports. It is the essential mechanism that separates network switches from network hubs.

Attacker may spoof fake Ethernet frames and overload CAM table of switch. Due to this, Switch performance rapidly decreases and switch may start to flood some traffic as hub. Port-security which limits a number of learned MAC addresses is the countermeasure against this type o attack.

We will successfully test the attack first and then apply port-fast on the switch and see how it works.

TOPOLOGY



SCENARIO

1. Initial setup of topology on *Switch*
2. Initial setup of topology on *VPC_Victims*
3. Perform CAM overflow attack
4. Deploy PortSecurity
5. Verify PortSecurity operation

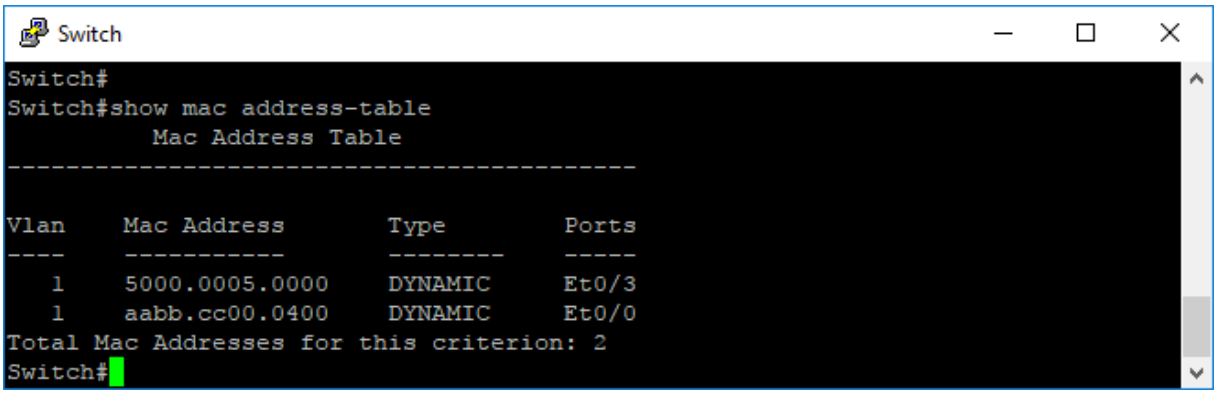
INITIAL SETUP OF TOPOLOGY ON SWITCH

1. *Switch* port must be access port connecting end-hosts

```
Switch> enable
Switch# configure terminal
Switch(config)# interface range e0/0-3
Switch(config-if)# switchport mode access
```

2. Check the content of CAM table

```
Switch(config-if)# end
Switch# show mac address-table
```



INITIAL SETUP OF TOPOLOGY ON VICTIMS

1. Assign IP address to *VPC_Victim1*

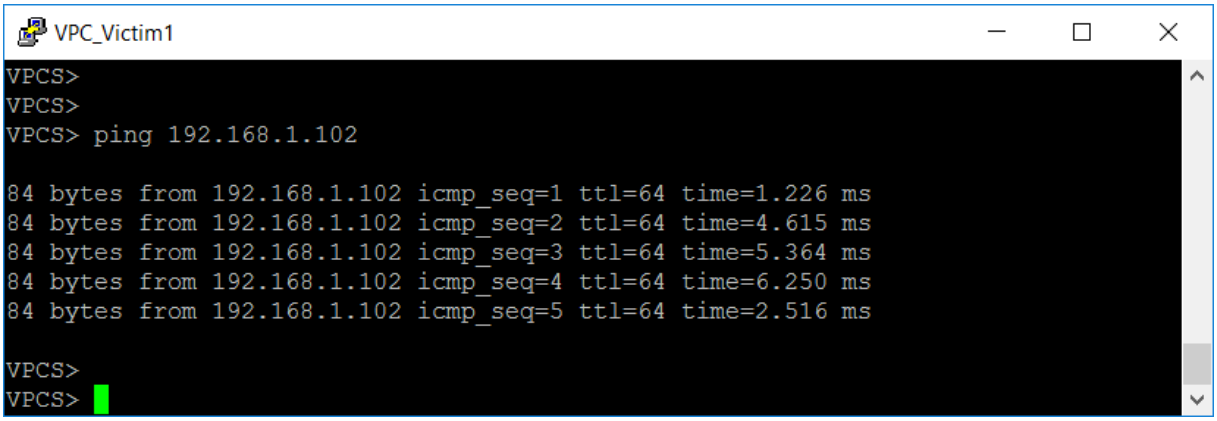
   VPCS> **ip 192.168.1.101/24 192.168.1.1**

2. Assign IP address to *VPC_Victim2*

   VPCS> **ip 192.168.1.102/24 192.168.1.1**

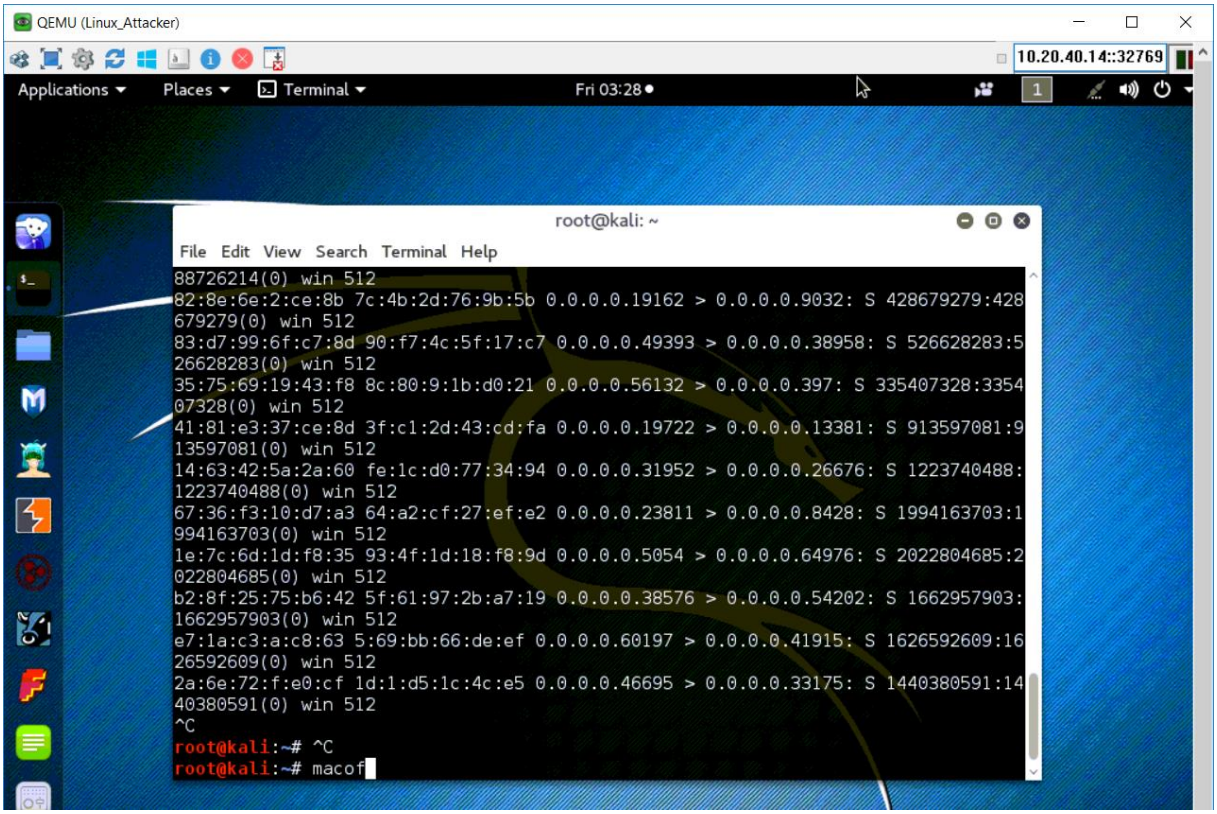
3. Try to ping between victims

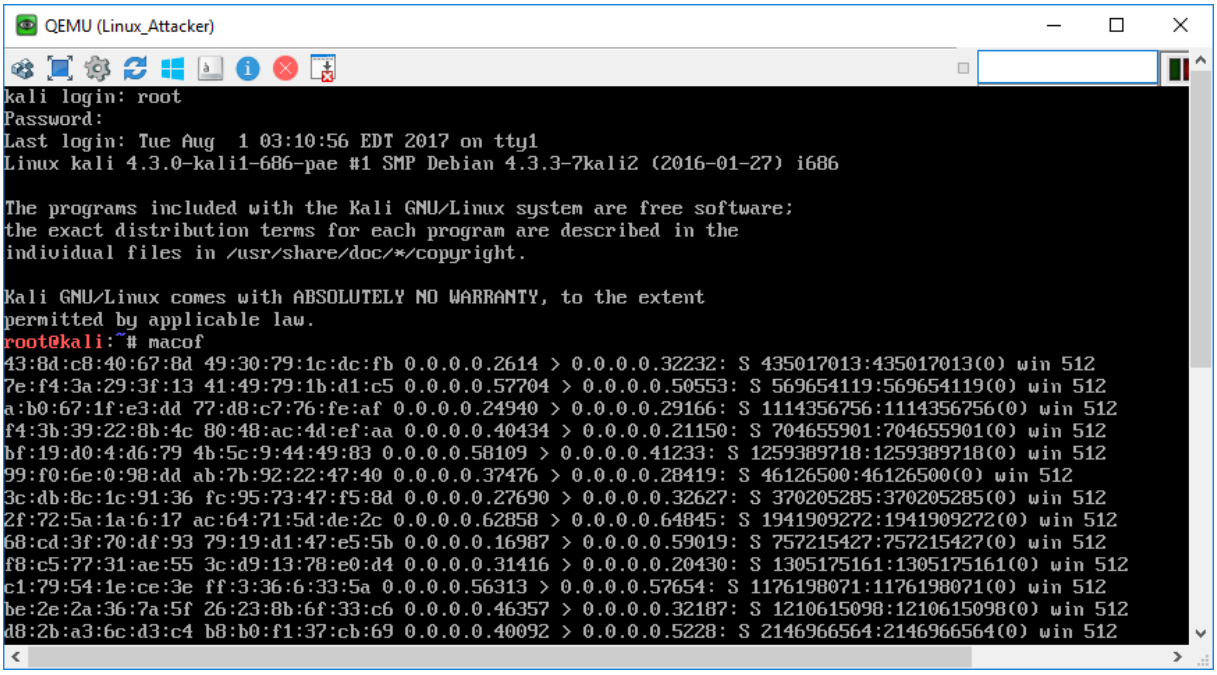
   VPCS> **ping 192.168.1.102**



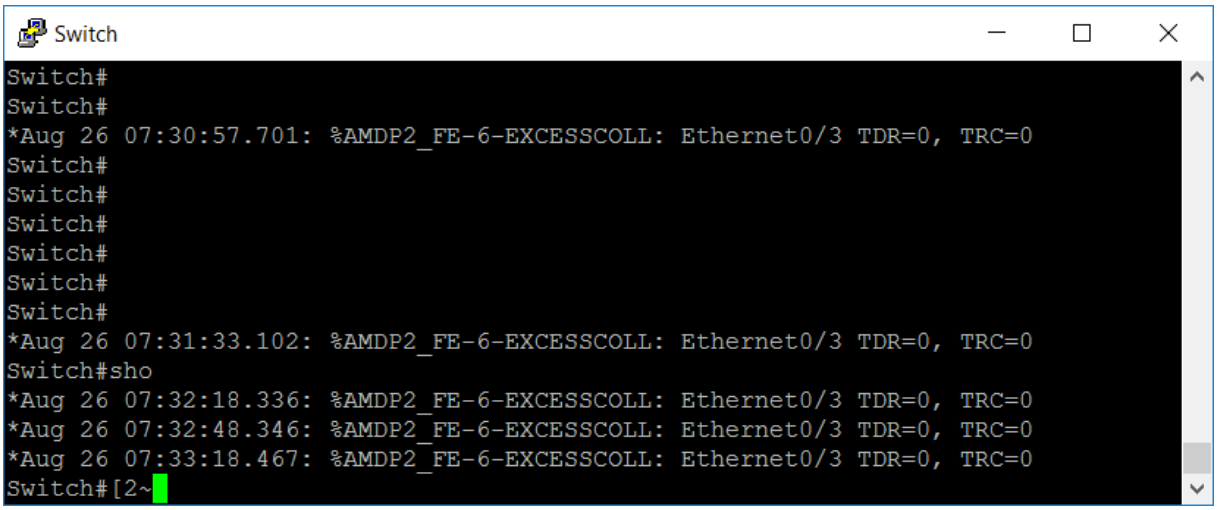
PERFORM CAM OVERFLOW ATTACK

1. First logon on to attacker's computer, username "root", password "toor"
    a. Use GUI
    b. Alternatively, use CLI by pressing Ctrl+Alt+F1
2. Open terminal
3. Execute program **`macof`**

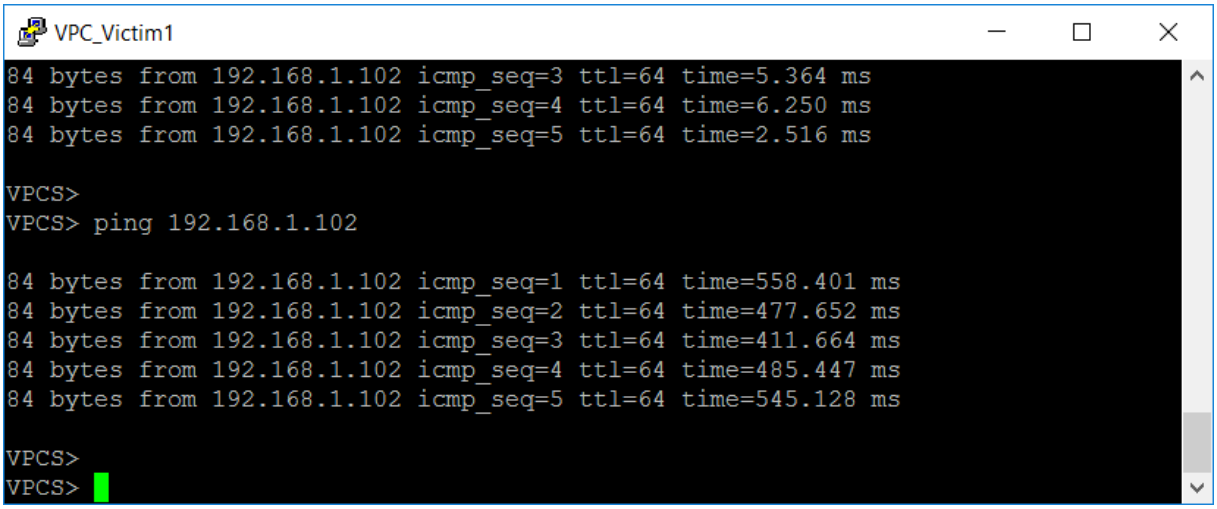




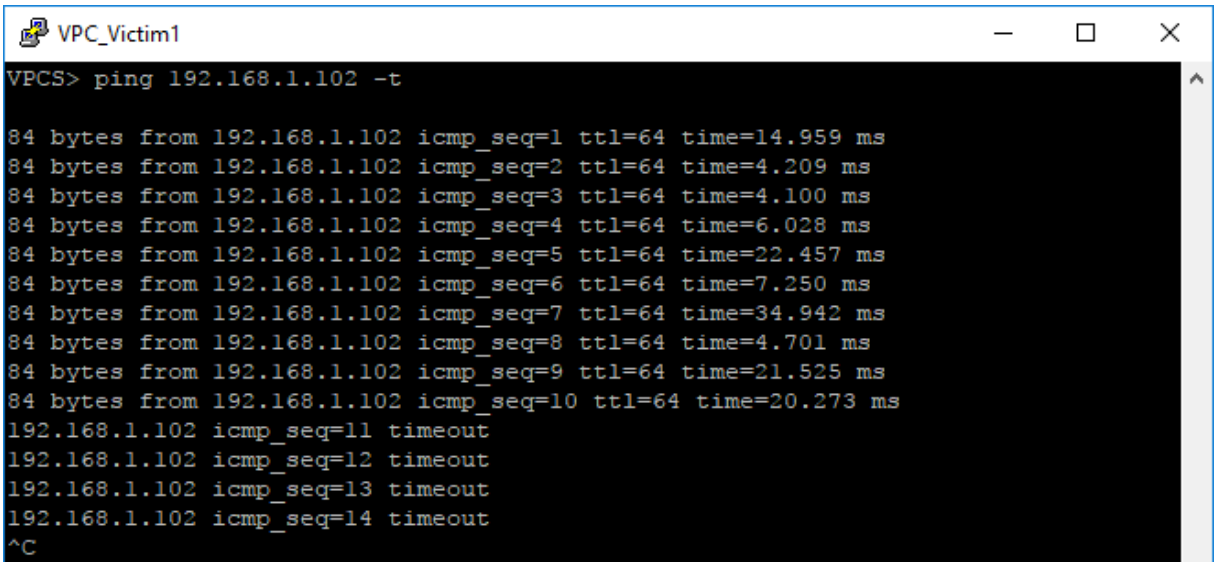
4. Attacker's computer starts to spoof nonsentic Ethernet frames to various addresses and *Switch*'s CPU overloads (you can issue any command on Switch)



5. Observe the ping time between victims increases

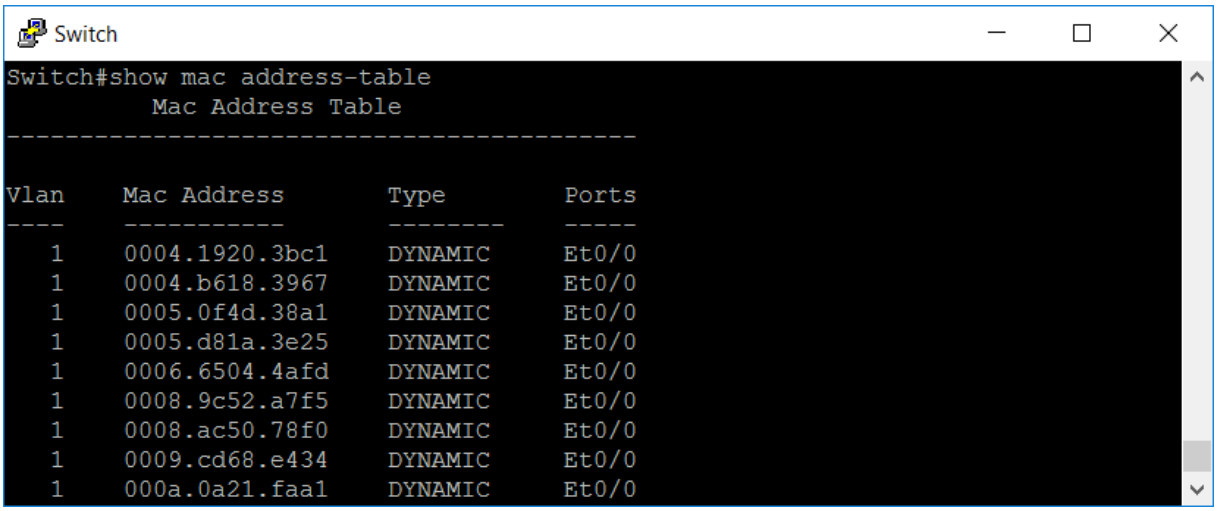


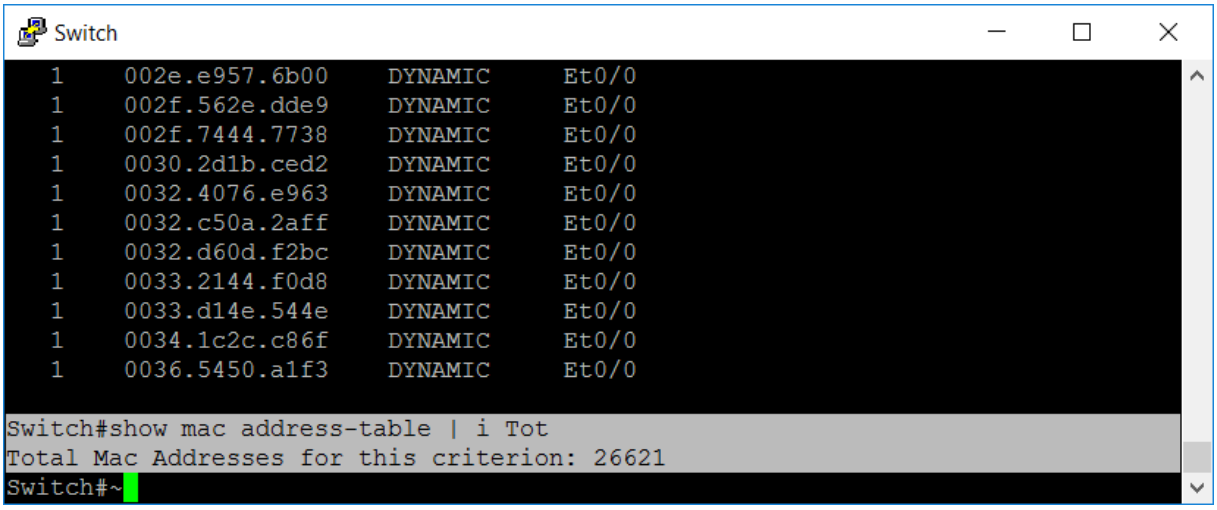
If you will keep attack running for a while, pings might even get lost



6. Cancel attack by typing Ctrl+C on attacker and show *Switch*'s CAM table

```
Switch# show mac address-table
```





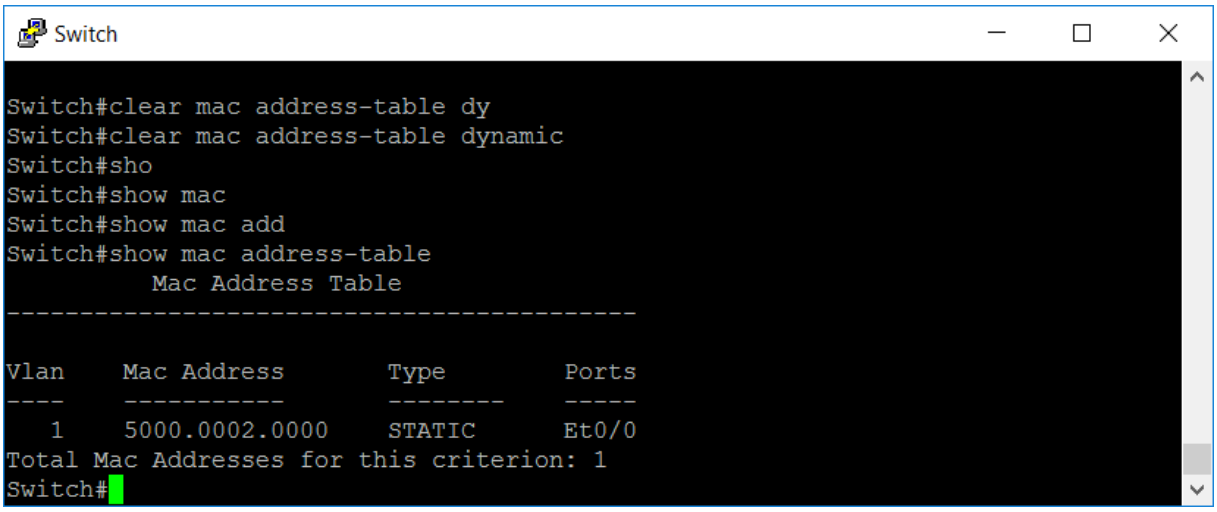
DEPLOY PORTSECURITY ON *SWITCH*

1. Configure it on interfaces

```
Switch(config)# interface range e0/0-3
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security violation shutdown
Switch(config-if)# switchport port-security maximum 2
Switch(config-if)# end
```

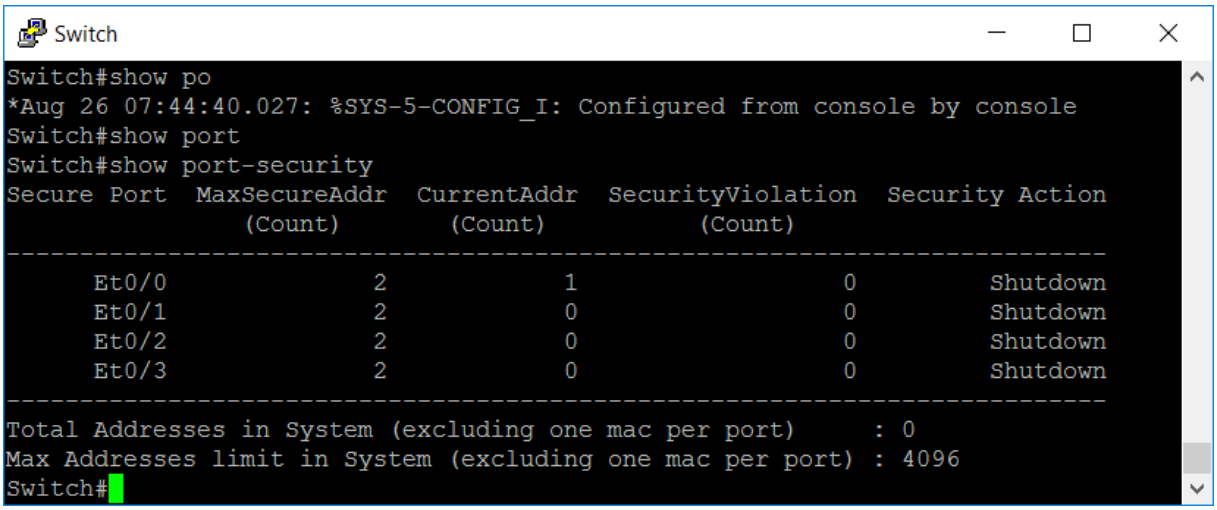
2. Clear currently full CAM table and verify

```
Switch# clear mac address-table dynamic
Switch# show mac address-table
```



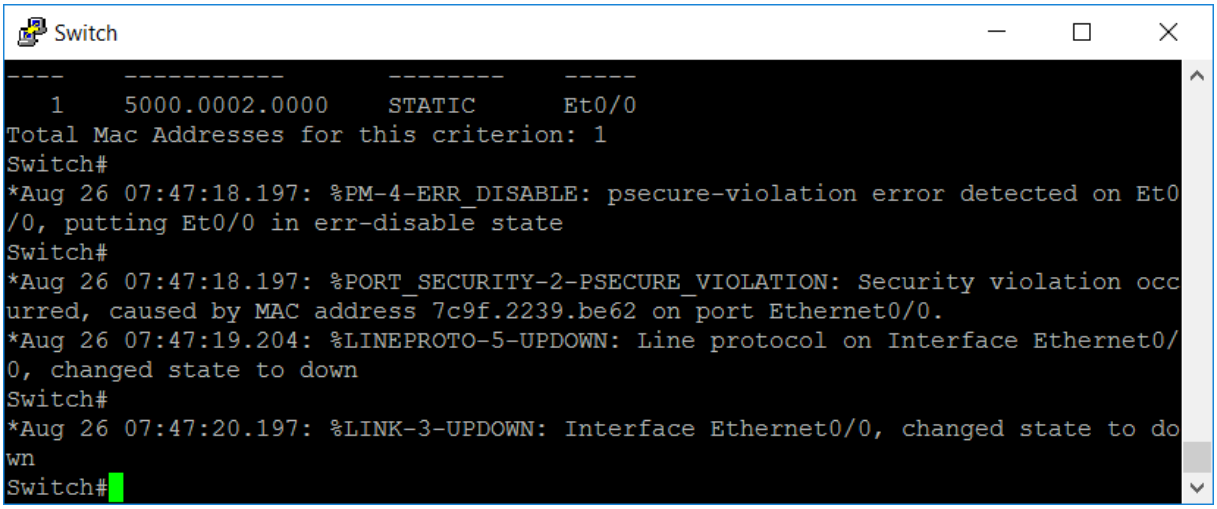
3. Verify configuration

```
Switch# show port-security
```

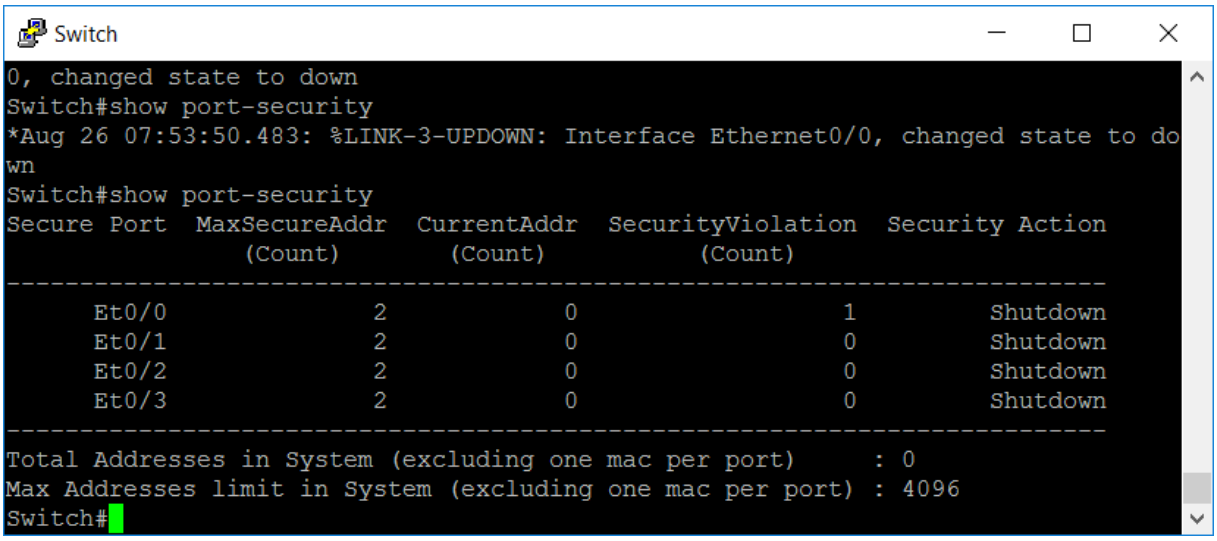


VERIFY PORTSECURITY OPERATION

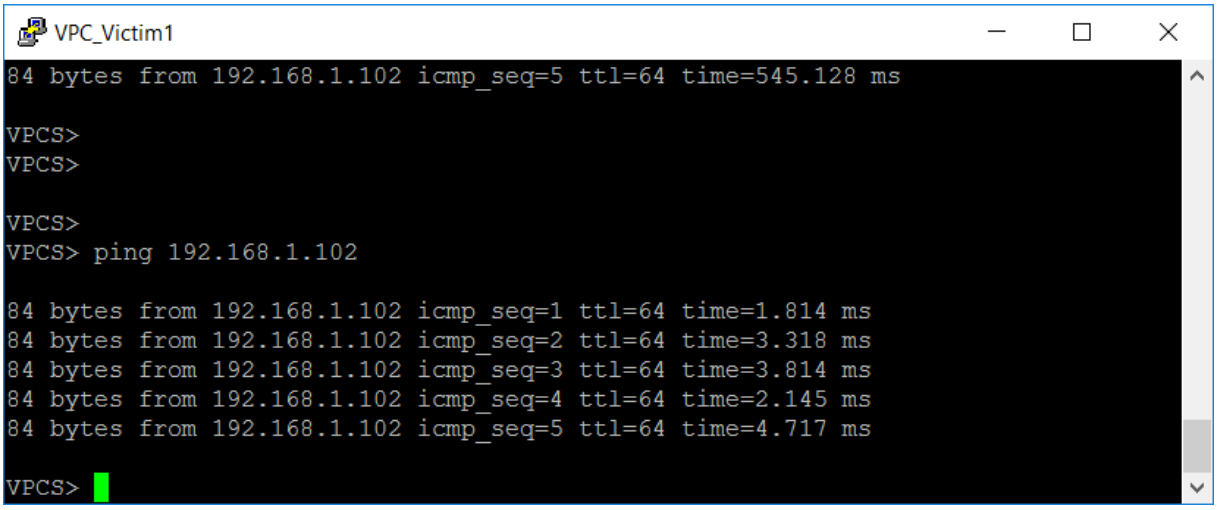
1. Once again issue **`macof`** in attacker's terminal
2. Switch immediately shutsdown attacker's port and attacker is isolated from the network. Observe syslog messages notifying about this event.



3. Number of port-security violation has increased



4. Traffic between victims flows uninterrupted



5. Port is in error-disabled mode and administrator must manually reenable interface in order to make it operational.

```
Switch# show interface status err-disable
Switch# configure terminal
Switch(config)# interface e0/0
Switch(config-if)# shutdown
Switch(config-if)# no shutdown
```

